

Matrices in Geometry 4.3.26

EE25BTECH11037 - Divyansh

Question: Find the ratio in which the line segment joining the points $\mathbf{A} = (4, 8, 10)$ and $\mathbf{B} = (6, 10, -8)$ is divided by the YZ plane.

Solution: We have two points $\mathbf{A} = \begin{pmatrix} 4 \\ 8 \\ 10 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 6 \\ 10 \\ -8 \end{pmatrix}$

The direction vector $\mathbf{d}$ is given by

$$\mathbf{d} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} 2 \\ 2 \\ -18 \end{pmatrix} \quad (1)$$

Writing the parametric equation of a point $\mathbf{P}$ on the line joining $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{P} = \mathbf{A} + t\mathbf{d} \quad (2)$$

$$\Rightarrow \mathbf{P} = \begin{pmatrix} 4 \\ 8 \\ 10 \end{pmatrix} + t \begin{pmatrix} 2 \\ 2 \\ -18 \end{pmatrix} \quad (3)$$

$$\Rightarrow \mathbf{P} = \begin{pmatrix} 4 + 2t \\ 8 + 2t \\ 10 - 18t \end{pmatrix} \quad (4)$$

Let $\mathbf{P}$ be the point that intersects the Y-Z plane. Therefore, the x-coordinate of the point $\mathbf{P}$ is zero.

$$4 + 2t = 0 \Rightarrow t = -2 \quad (5)$$

$$\therefore \mathbf{P} = \begin{pmatrix} 0 \\ 4 \\ -26 \end{pmatrix} \quad (6)$$

Let $\mathbf{P}$ divide $\mathbf{A}$ and $\mathbf{B}$ in the ratio $k : 1$

Using the formula (1.1.5.2)

$$k = \frac{(\mathbf{A} - \mathbf{P})^\top (\mathbf{P} - \mathbf{B})}{\|\mathbf{P} - \mathbf{B}\|^2} \quad (7)$$

$$\Rightarrow k = \frac{\begin{pmatrix} 4 & 4 & -36 \end{pmatrix} \begin{pmatrix} -6 \\ -6 \\ 54 \end{pmatrix}}{\left\| \begin{pmatrix} -6 \\ -6 \\ 54 \end{pmatrix} \right\|^2} \quad (8)$$

$$\Rightarrow k = \frac{-24 - 24 - 1944}{36 + 36 + 2916} = \frac{-1992}{2988} \quad (9)$$

$$\Rightarrow k = \frac{-2}{3} \quad (10)$$

Hence, the Y-Z plane divides the line segment that joins the points $\mathbf{A}$ and $\mathbf{B}$ in the external ratio $2 : 3$.

3D Plot: Points A, B, P, Line AB, YZ-Plane, and Connections

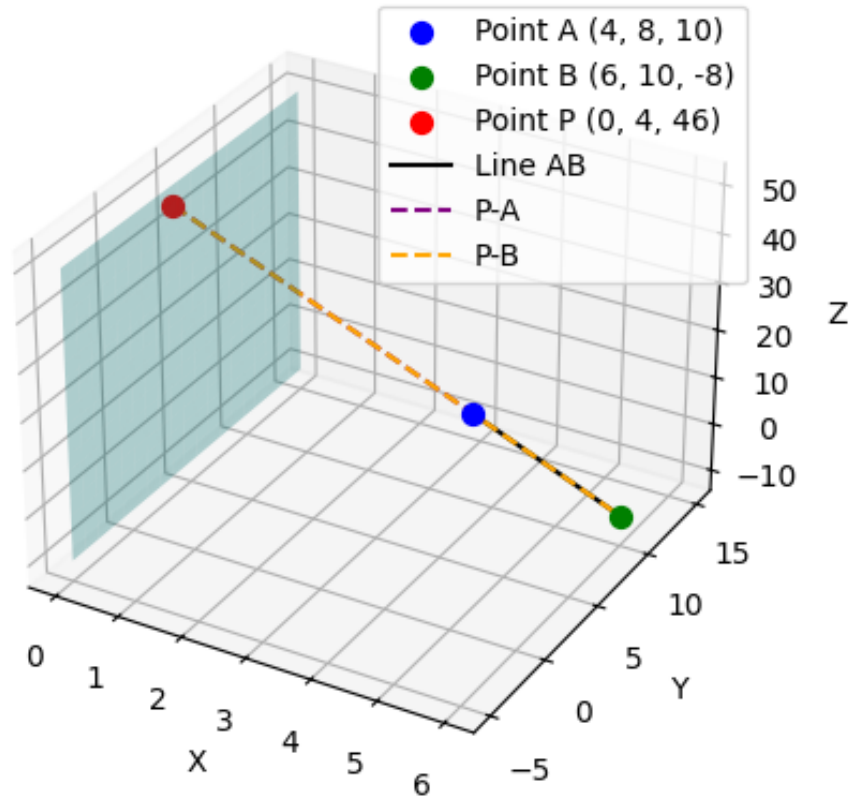


Fig. 1: Graph for 4.3.26